

Car Crash Prediction on Busy Streets Using Logistic Regression and XGBoost

Abstract

Road traffic accidents remain a major public safety concern, particularly in densely populated urban areas. This study develops a machine learning framework to predict the likelihood of a car crash occurring on a busy street using traffic, environmental, and infrastructure-related variables. A synthetic dataset consisting of 6,500 observations was generated using a hidden risk formulation based on domain-inspired relationships between traffic conditions and crash occurrence. Two machine learning models, Logistic Regression and XGBoost, were trained and evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC metrics. Feature importance and multivariate odds ratio analyses were performed to identify the most influential crash predictors. Threshold optimization was conducted using Precision-Recall-F1 analysis, and DeLong's test was used to statistically compare model performance. Results showed that Logistic Regression achieved the highest ROC-AUC score of 0.854 and significantly outperformed XGBoost (ROC-AUC = 0.845, $p = 0.027$).

1. Introduction

Road crashes are influenced by multiple factors including traffic density, weather conditions, road infrastructure, visibility, pedestrian activity, and vehicle composition.

The objective of this study is to develop and compare machine learning models capable of predicting crash occurrence using traffic and environmental variables. The study focuses on model interpretability, statistical significance testing, and threshold optimization for decision-making.

2. Dataset Generation

A synthetic dataset containing 6,500 observations was generated.

Input Features

1. Traffic Density
2. Speed
3. Road Condition

4. Rainfall
5. Signal Delay
6. Road Intersections
7. Visibility
8. Pedestrian Count
9. Time of Day
10. Vehicle Type Mix

Target Variable

Crash Occurs:

- 1 = Crash
- 0 = No Crash

A hidden risk score was generated using weighted relationships among the predictors. The risk score was transformed into crash probabilities using the logistic sigmoid function, after which crash labels were generated using Bernoulli sampling.

The final dataset was nearly balanced:

- Crash = 49.69%
 - No Crash = 50.31%
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3. Data Preprocessing

The following preprocessing steps were performed:

- Missing value verification
- One-hot encoding of categorical variables
- Train-test split (80%-20%)
- Feature scaling for Logistic Regression using StandardScaler

Training samples: 5,200

Testing samples: 1,300

4. Logistic Regression Model

Logistic Regression was selected due to its interpretability and ability to provide multivariate odds ratios.

Performance Metrics

Metric	Value
Accuracy	0.767
Precision	0.765
Recall	0.766
F1-Score	0.766
ROC-AUC	0.854

The ROC curve demonstrated strong discrimination capability with an AUC of 0.854.

5. Multivariate Odds Ratio Analysis

Odds ratios were calculated using:

$$OR = \exp(\beta)$$

where β represents the Logistic Regression coefficient.

Major Risk Factors

Feature	Odds Ratio
Rainfall	3.13
Traffic Density	1.80
Pedestrian Count	1.78
Speed	1.56

Interpretation:

- Rainfall increases crash odds by approximately 213%.
- High traffic density increases crash odds by 80%.
- Increased pedestrian activity increases crash odds by 78%.
- Higher speeds increase crash odds by 56%.

Protective Factors

Feature	Odds Ratio
Road Condition	0.41
Visibility	0.82

Interpretation:

- Better road conditions reduce crash odds by approximately 59%.
 - Improved visibility reduces crash odds by approximately 18%.
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6. XGBoost Model

XGBoost was implemented to capture potential nonlinear relationships among the predictors.

Performance Metrics

Metric	Value
Accuracy	0.765
Precision	0.765
Recall	0.760
F1-Score	0.762
ROC-AUC	0.845

Although XGBoost performed competitively, its performance was slightly lower than Logistic Regression.

7. Feature Importance Analysis

XGBoost feature importance analysis identified the following variables as the most influential:

1. Road Condition
2. Rainfall
3. Pedestrian Count
4. Traffic Density
5. Speed

These findings were consistent with the odds ratio analysis performed using Logistic Regression.

8. Threshold Optimization

The default classification threshold of 0.50 was compared against multiple threshold values ranging from 0.10 to 0.90.

The highest F1-score was achieved at:

Threshold = 0.40

Threshold	Precision	Recall	F1
0.40	0.728	0.858	0.788
0.50	0.765	0.766	0.766

Although Precision and Recall were nearly equal at 0.50, the threshold of 0.40 provided a higher F1-score and substantially improved Recall.

This is particularly important for crash prediction systems, where failing to detect a crash event (False Negative) is generally more costly than issuing an additional warning (False Positive).

Therefore, a threshold of 0.40 was selected as the optimal operating point.

9. DeLong Test

A DeLong test was conducted to compare the ROC-AUC values of Logistic Regression and XGBoost.

Results

Metric	Value
Logistic Regression AUC	0.8536
XGBoost AUC	0.8451
Z Statistic	2.210
P-value	0.0271

Hypothesis

H_0 : $AUC(\text{Logistic Regression}) = AUC(\text{XGBoost})$

H_1 : $AUC(\text{Logistic Regression}) \neq AUC(\text{XGBoost})$

Since:

$$p = 0.0271 < 0.05$$

the null hypothesis was rejected.

This indicates that Logistic Regression achieved statistically superior discriminatory performance compared to XGBoost.

10. Discussion

The synthetic crash generation process was constructed using a linear risk formulation followed by a logistic transformation. Consequently, Logistic Regression closely matched the underlying data-generating mechanism and achieved superior predictive performance.

Rainfall, traffic density, pedestrian activity, and speed emerged as the dominant crash risk factors. Conversely, improved road conditions and visibility significantly reduced crash risk.

Both Logistic Regression and XGBoost identified similar risk patterns, demonstrating consistency and reliability in feature interpretation.

11. Conclusion

This study developed a machine learning framework for predicting crash occurrence on busy streets using traffic and environmental variables. Logistic Regression achieved the highest predictive performance with an ROC-AUC of 0.854 and statistically outperformed XGBoost according to DeLong's test ($p = 0.0271$). Threshold optimization identified 0.40 as the optimal operating threshold, providing the best balance between Precision and Recall. The findings highlight the importance of rainfall, traffic density, pedestrian activity, and road conditions in influencing crash risk and demonstrate the value of interpretable machine learning techniques for transportation safety applications.

12. References

References

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